Big Data

BY THE END:

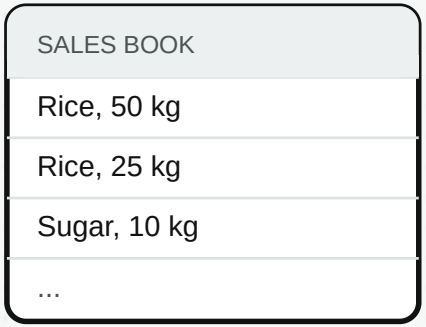
You can say what makes data "big", and judge whether a dataset actually is.

Overview

A market trader's sales book holds four hundred rows and answers every question the trader has. A digital wallet serving millions of customers records four hundred payments while you read this sentence. Both are data, collected the same way, describing the same kind of event. Only one of them breaks an ordinary spreadsheet, and the difference is not what the data means. It is how much of it arrives, how fast, and in how many shapes.

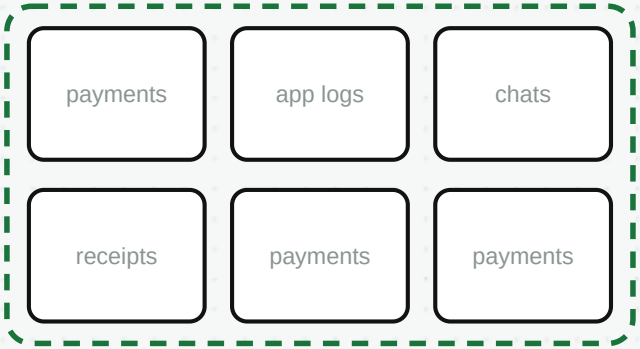
The trader and the wallet

A market trader



400 rows, one table
a few sales an hour

A digital wallet



billions of records, many shapes
thousands a second

same kind of fact, different problem

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What makes data big

Big data is data at a scale that ordinary tools and methods can no longer handle. There is no row count that makes the label true. What matters is whether the usual approach has stopped working.

THE THREE VS	THE QUESTION IT ANSWERS	THE MARKET TRADER	THE DIGITAL WALLET
Volume	How much is there?	400 rows in a book	Billions of payment records
Velocity	How fast does it arrive?	A few sales an hour	Thousands of payments a second
Variety	How many shapes does it come in?	One table	Payments, app logs, support chats, photos of receipts

Variety you have already met. It is the structured, semi-structured, and unstructured split from the types lesson, seen from a distance.

Big is a relationship, not a number

The same dataset can be big for one organization and ordinary for another. A hospital that has always worked in spreadsheets meets big data at a few million rows. A payment company meets it much later, because it already owns the systems.

> The honest test is a question about the tools, not the data: **has the usual way of working broken?** If a file still opens, a query still returns, and one machine still holds it, the dataset is not big, whatever its subject.

Most data work is not big data work. Calling a small dataset "big data" is how organizations end up buying systems they never needed, to solve a problem they did not have.

What changes when data really is big

Storage spreads out. The data sits across many machines instead of one, which is what a data lake is for.

Errors hide. In four hundred rows you can see a bad value. In four billion you have to detect it, and anything you cannot detect you inherit.

Deletion gets hard. Finding every copy of one person's records across many systems is real work, which is why retention rules are written before the data piles up.

Someone has to build the plumbing. Moving data at this scale is a full-time job, and a field exists around it.

Big is not the same as useful

A large dataset is not automatically a better one. Everything from the quality lesson still applies, and applies harder: billions of rows collected carelessly are billions of rows of careless data, and the scale makes the flaw more convincing rather than less.

> The trader with four hundred honest rows can answer the trader's question. **That is worth more than four billion rows nobody checked.**